

Mathematical Notes for uwnav_dynamics

Project uwnav_dynamics

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1 Notation

1.1 Discrete Time and Windows

Let the aligned main timeline be indexed by $t \in \mathbb{Z}$ with sampling interval Δt . In the current implementation, $\Delta t = 0.01$ s, corresponding to a 100 Hz main axis.

We use:

- L : history window length,
- H : prediction horizon length.

1.2 Control, Observation, and Supervision

At time t , the control input is denoted by

$$\mathbf{u}_t \in \mathbb{R}^8, \quad (1)$$

corresponding to the eight PWM-related thruster command channels.

The aligned observation feature vector is denoted by

$$\mathbf{z}_t \in \mathbb{R}^{D_{\text{in}}}, \quad (2)$$

where the current baseline uses $D_{\text{in}} = 25$ and collects PWM, IMU, velocity-state, and Power features.

The supervised prediction target is denoted by

$$\mathbf{y}_t \in \mathbb{R}^{D_{\text{out}}}, \quad (3)$$

where the current baseline uses $D_{\text{out}} = 9$ and groups acceleration, angular-rate, and velocity-state terms.

The history input to the model is

$$\mathbf{X}_t = \begin{bmatrix} \mathbf{z}_{t-L+1} \\ \vdots \\ \mathbf{z}_t \end{bmatrix} \in \mathbb{R}^{L \times D_{\text{in}}}. \quad (4)$$

The future target window is

$$\mathbf{Y}_{t+1:t+H} = \begin{bmatrix} \mathbf{y}_{t+1} \\ \vdots \\ \mathbf{y}_{t+H} \end{bmatrix} \in \mathbb{R}^{H \times D_{\text{out}}}. \quad (5)$$

1.3 Prediction Outputs

The model predicts

$$\Delta \hat{\mathbf{Y}}_{t+1:t+H} \in \mathbb{R}^{H \times D_{\text{out}}}, \quad (6)$$

as well as diagonal log-variance terms

$$\log \hat{\boldsymbol{\sigma}}_{t+1:t+H}^2 \in \mathbb{R}^{H \times D_{\text{out}}}. \quad (7)$$

Rollout uses the last available state proxy $\mathbf{y}_t$ as initial condition and reconstructs future states from cumulative increments.

2 Underwater Dynamics Model and Motivation

2.1 Classical Dynamics Background

A standard underwater vehicle dynamics model may be written as

$$\mathbf{M}(\boldsymbol{\nu})\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{g}(\boldsymbol{\eta}) = \boldsymbol{\tau}, \quad (8)$$

where $\boldsymbol{\nu}$ denotes body-frame velocity variables, $\boldsymbol{\eta}$ denotes pose variables, and $\boldsymbol{\tau}$ denotes generalized actuation.

This form is physically meaningful but difficult to identify precisely in practice because:

- thruster dynamics are nonlinear and history-dependent,
- hydrodynamic effects vary with operating condition,
- sensor observations are multi-rate and partially missing,
- the final use case is short-horizon prediction for control rather than full parameter recovery.

2.2 Why a Data-Driven Surrogate

Instead of identifying all coefficients explicitly, the current project adopts a control-oriented surrogate:

$$\mathbf{y}_{t+1:t+H} \approx f_{\theta}(\mathbf{X}_t). \quad (9)$$

The model is therefore asked to approximate the locally relevant short-horizon response map from aligned history windows to future observable state proxies.

2.3 Structured Priors

The current architecture keeps room for weak physical priors:

- actuator-side lag and saturation modeling,
- latent hydrodynamic memory,
- damping-related output bias,
- heteroscedastic predictive uncertainty.

These priors do not replace the classical dynamics equation; they provide an intermediate layer between pure black-box prediction and full white-box identification.

3 Neural Dynamics Model

3.1 Sequence-to-Horizon Prediction

The current neural predictor maps a history window to future increments:

$$\left(\Delta \hat{\mathbf{Y}}_{t+1:t+H}, \log \hat{\boldsymbol{\sigma}}_{t+1:t+H}^2 \right) = f_{\theta}(\mathbf{X}_t). \quad (10)$$

The baseline implementation uses an LSTM encoder followed by prediction heads.

3.2 Rollout Semantics

Let $\mathbf{y}_t$ denote the last available target-aligned state proxy extracted from the input window. The predicted future sequence is reconstructed as

$$\hat{\mathbf{y}}_{t+j} = \mathbf{y}_t + \sum_{k=1}^j \Delta \hat{\mathbf{y}}_{t+k}, \quad j = 1, \dots, H. \quad (11)$$

This is the “delta-cumsum” rollout used by the current training and evaluation pipelines.

3.3 Optional Structured Modules

The architecture may include the following modules:

- **ThrusterLag**: transforms raw command history into an effective control sequence,
- **HydroSSM**: provides a latent hydrodynamic memory state,
- **DampingHead**: adds a damping-related prior to the predicted increments,
- **UncertaintyHead**: predicts diagonal log-variance terms.

Accordingly, one may interpret the network abstractly as

$$(\Delta \hat{\mathbf{Y}}, \log \hat{\boldsymbol{\sigma}}^2) = f_{\theta}(\mathbf{X}_t, \mathbf{h}_t), \quad (12)$$

where $\mathbf{h}_t$ denotes optional latent structured features induced by the additional modules.

4 Training Objective

4.1 Current Baseline Objective

The current baseline predicts both mean trajectory increments and diagonal uncertainty. After rollout, the baseline objective is a diagonal Gaussian negative log-likelihood:

$$\mathcal{L}_{\text{NLL}} = \sum_{j=1}^H \sum_{d=1}^{D_{\text{out}}} \left[\frac{(y_{t+j,d} - \hat{y}_{t+j,d})^2}{\hat{\sigma}_{t+j,d}^2} + \log \hat{\sigma}_{t+j,d}^2 \right]. \quad (13)$$

4.2 Dense and Sparse Supervision

The engineering challenge is that supervision is not uniformly dense across modalities:

- IMU-related target components are available on the main aligned axis,
- DVL-related velocity information is lower-rate and naturally sparse.

This motivates a mask-aware formulation:

$$\mathcal{L}_{\text{mask}} = \sum_{j=1}^H \sum_{d=1}^{D_{\text{out}}} m_{t+j,d} \left[\frac{(y_{t+j,d} - \hat{y}_{t+j,d})^2}{\hat{\sigma}_{t+j,d}^2} + \log \hat{\sigma}_{t+j,d}^2 \right], \quad (14)$$

where $m_{t+j,d} \in \{0, 1\}$ encodes whether the target dimension is valid at that time step.

4.3 Interpretation

In the current repository, the dense baseline is already implemented, while the fully mask-aware loss is a planned extension. The mathematical form above is therefore both:

- a precise statement of the intended sparse-supervision objective,
- a bridge between the current baseline and future mask-aware training.

5 Control Integration

5.1 Model Interface for Control

The long-term target is to use the learned predictor as a short-horizon dynamics model inside a controller:

$$\hat{\mathbf{y}}_{t+1:t+H} = f_{\theta}(\mathbf{X}_t, \mathbf{u}_{t:t+H-1}), \quad (15)$$

possibly together with uncertainty outputs.

5.2 MPC-Oriented Objective

A generic finite-horizon control objective can be written as

$$\min_{\mathbf{u}_{t:t+H-1}} \sum_{k=0}^{H-1} (\|\hat{\mathbf{x}}_{t+k} - \mathbf{x}_{t+k}^*\|_{\mathbf{Q}}^2 + \|\mathbf{u}_{t+k}\|_{\mathbf{R}}^2), \quad (16)$$

subject to actuator and safety constraints.

5.3 Current Status

The repository is not yet an MPC implementation. However, the current training and evaluation design already prepares the ingredients needed for future integration:

- short-horizon rollout semantics,
- uncertainty outputs,
- configuration and artifact traceability,
- explicit room for structured priors.

5.4 Near-Term Research Direction

Before closed-loop integration, the following steps are still required:

- mask-aware sparse supervision for low-rate observations,
- stronger rollout contract consistency,
- clearer evaluation metrics that reflect control utility.